

Summary

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$$f: \mathbb{R}^5 \times \mathbb{R}^{33600 \times 3} \rightarrow \mathbb{R}^{33600}$$

$$y_i = f(\theta, U)_i = s(U_i) + \sum_{j \in N(i)} W_{ij}(\theta) s(U_j) + \epsilon_i$$

θ global settings (colour, coverage) $\mathbb{R}^5$

U per nozzle input field $\mathbb{R}^{33600 \times 3}$

$U_i = (V_{h(i)}, Fr_{h(i)}, dt2_i)$ input vector for nozzle i $\mathbb{R}^3$
 $h(i) \in \{1 \dots 30\}$

$s(U_i)$ response of nozzle i $s: \mathbb{R}^3 \rightarrow \mathbb{R}$

$N(i)$ spatial neighbours of nozzle i tbd

$W_{ij}(\theta)$ coupling weight from nozzle j to i tbd

ϵ_i noise $\mathbb{R}$

y_i measured SD valve at nozzle i $\mathbb{R}$

in vector form

$$y = f(\theta, U) = s(U) + W(\theta) s(U) + \epsilon$$

y full output vector $\mathbb{R}^{33600}$

$s(U)$ input settings for all nozzles $\mathbb{R}^{33600}$

$W(\theta)$ full weight matrix $\mathbb{R}^{33600} \times \mathbb{R}^{33600}$

ϵ noise (i know) $\mathbb{R}^{33600}$

simplified:

$$f(\theta, U) = (I + W(\theta)) s(U) + \epsilon$$

to be defined:

- Functional form of s : which model maps $U_i \in \mathbb{R}^3$ to a scalar response?
- Neighbourhood $N(i)$: how many neighbours and by what criterion?
- Functional form of $W_{ij}(\theta)$: data driven? does it really depend on θ ?

what I still want to do:

define space as continuous. This allows for the use of spatial kernels

$$W = (x, x', \theta)$$

defining how much influence x' has on position x .

Redefinition:

Let $X \subset \mathbb{R}^2$ be the physical print surface, then:

• $U: X \rightarrow \mathbb{R}^3$ — continuous control field

• $\gamma: X \rightarrow \mathbb{R}$ — continuous response field

and f is the operator mapping fields to fields

$$f: \mathbb{R}^3 \times (X \rightarrow \mathbb{R}^3) \rightarrow (X \rightarrow \mathbb{R})$$

since the 33600 samples are discrete values from

$$(x_j, U(x_j), \gamma(x_j)), \quad j \in \{1, \dots, 33600\}$$

the model becomes

$$\gamma(x) = s(U(x)) + \int_x W(x, x', \theta) s(U(x')) dx' + \epsilon(x)$$

with the discrete sum $\sum_j W_{ij}$ being the numerical approximation of this integral

let's clarify the integral first:

in the discrete case, for each nozzle i we sum the contributions of its neighbours:

$$\sum_{j \in N(i)} W_{ij} \cdot s(U_j)$$

if the nozzles are infinitely dense, the sum becomes an integral

$$\sum_j W(x, x') \cdot s(U(x')) \xrightarrow{\text{continuous}} \int_x W(x, x') \cdot s(U(x')) dx'$$

we are integrating over $x' \in X$, meaning all possible positions on the print surface.

with $W(x, x', \theta)$, the integral becomes finite ^{and defined}, because far away points contribute ~ 0 . so effectively:

$$\int_{N(x)} W(x, x') \cdot s(U(x')) dx'$$

where $N(x)$ is a "region" around x rather than an index set.

btw $s(U(x'))$ is a composition of 2 functions:

$U(x') \in \mathbb{R}^3 \rightarrow$ control field evaluated at position x' , or the input vector (V_h, F_{r_h}, dt_z) at that point

$s: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \rightarrow$ baseline nozzle response model